

The trapezium rule

Cambridge insert: what to do when the antiderivative does not exist — and how to judge the error.

LESSON 55–57

3 × 40 MIN

ALGEBRA 11, §3.2 (EXTENSION)

P2 · 5.7

LESSON CLOCK

18'

28'

28'

26'

30'

5

Why a numerical method	18 min
The rule	28 min
Over or under?	28 min
Improving the estimate	26 min
Practice	30 min
Homework	5 min

LEARNING OBJECTIVES

- State and apply the trapezium rule with n strips.
- Decide whether the estimate is an over- or an under-estimate.
- Improve an estimate by doubling the number of strips.
- Use the rule on data given only as a table.

TEXTBOOK

National	pp. 261–268
Cambridge	Pure Mathematics 2 & 3, pp. 108–116
Workbook	P2 Exercise 5E

01

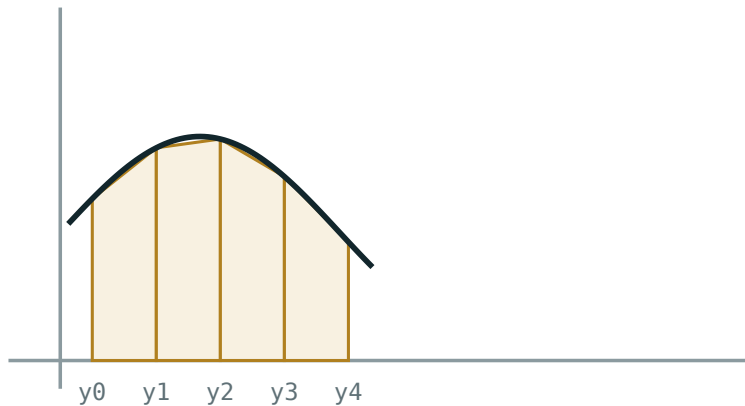
Explanation

Why a numerical method

Not every function has an antiderivative that can be written down. $\int \sqrt{1+x^3} \, dx$ and $\int 2^{\sqrt{x^2}} \, dx$ exist as areas but have no elementary formula. And data collected by measurement has no formula at all.

THE IDEA

Replace the curve on each strip by the **straight line** joining its ends. Each strip becomes a trapezium, whose area is elementary.



straight tops, not flat ones – the trapezium rule

Straight tops, not flat ones. Four strips already fit closely.

The rule

$$\int_a^b y \, dx \approx \frac{h}{2} [y_0 + y_n + 2(y_1 + y_2 + \dots + y_{(n-1)})]$$

with $h = \frac{b-a}{n}$. The **end** ordinates count once; every **interior** ordinate counts twice, because it is the right edge of one trapezium and the left edge of the next.

N STRIPS MEANS N + 1 ORDINATES

Four strips need five values of y . Counting the strips as ordinates is the commonest arithmetic error, and it makes h wrong as well.

Example. $\int_0^4 x^2 \, dx$ with 4 strips: $h = 1$, ordinates 0, 1, 4, 9, 16.

$$\frac{1}{2} [0 + 16 + 2(1 + 4 + 9)] = \frac{1}{2} (16 + 28) = 22$$

The exact value is $\frac{64}{3} \approx 21.33$. The estimate is high, and the next section says why that was predictable.

Over or under?

THE RULE OF THUMB

Where the curve is **concave up** (bending upwards, $f'' > 0$), the chord lies **above** the curve, so the trapezium rule **over-estimates**. Where it is concave down, it under-estimates.

$y = x^2$ is concave up everywhere, so $22 > 21.33$ — exactly as predicted. This is a real mark in the examination: “state, with a reason, whether your estimate is an over- or an under-estimate.”

CURVE ON $[A, B]$	SHAPE	ESTIMATE
$y = x^2$	concave up	over
$y = \sqrt{x}$	concave down	under
$y = \frac{1}{x}, x > 0$	concave up	over
$y = x^3, x > 0$	concave up	over

Improving the estimate

Halving h — doubling the number of strips — divides the error by about **four**. The error is proportional to h^2 .

STRIPS	H	ESTIMATE OF $\int_0^4 x^2 dx$	ERROR
2	2	24	2.67
4	1	22	0.67
8	0.5	21.5	0.17

Each doubling quarters the error — visible in the last column. The pattern is what tells a computer when to stop.

From a table. When the data is measured rather than given by a formula, the rule applies unchanged: it needs only equally spaced ordinates, and never the formula itself.

02 Worked examples

EXAMPLE 1 Estimate $\int_0^4 x^2 dx$ with 4 strips.

① $h = 1$; ordinates $0, 1, 4, 9, 16$.

Five of them.

② $\frac{1}{2}[0 + 16 + 2(1 + 4 + 9)]$

③ $= \frac{1}{2}(44) = 22$

④ Concave up, so an over-estimate.

Exact 21.33.

ANSWER

22, an over-estimate

EXAMPLE 2 Estimate $\int_1^3 \frac{1}{x} dx$ with 4 strips.

① $h = 0.5$; ordinates $1, \frac{2}{3}, 0.5, 0.4, \frac{1}{3}$.

② $\frac{0.5}{2}[1 + 0.3333 + 2(0.6667 + 0.5 + 0.4)]$

③ $= 0.25 \times 4.4667$

④ ≈ 1.117

Exact $\ln 3 \approx 1.0986$.

ANSWER

≈ 1.117 , an over-estimate

EXAMPLE 3 A car's speed is measured every 10 s: 0, 8, 14, 18, 20 m/s. Estimate the distance.

① $h = 10$, 4 strips, 5 ordinates.

② $5[0 + 20 + 2(8 + 14 + 18)]$

③ $= 5(20 + 80)$

④ $= 500 \text{ m.}$

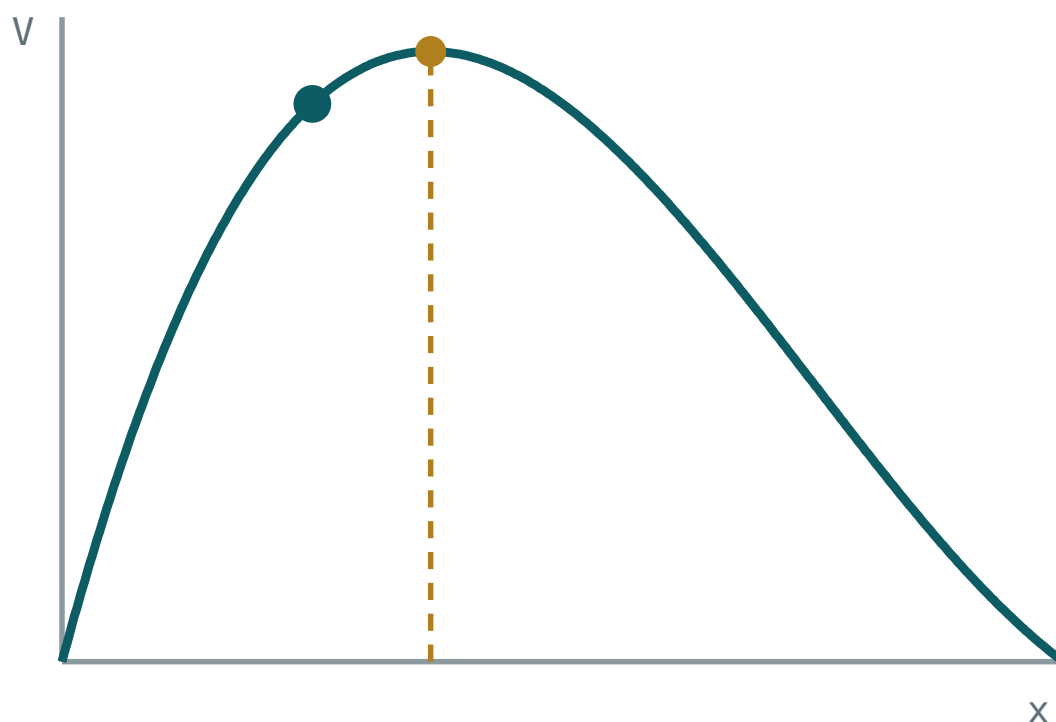
ANSWER

500 m

03 Interactive model

Compute the same integral with 2, 4 and 8 strips and put the errors side by side.

Strips under a curve


 $x = 2$
 $V(x) = 384$

greatest $V = 420.1$

at $x = 2.94$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Trapezium rule	Trapetsiya usuli	Метод трапеций
Strip	Bo'lak	Полоса
Ordinate	Ordinata	Ордината
Strip width h	Bo'lak kengligi	Шаг
Over-estimate	Ortiqcha baho	Оценка с избытком
Under-estimate	Kam baho	Оценка с недостатком
Concave up	Yuqoriga qavariq	Выпуклая вниз
Numerical integration	Sonli integrallash	Численное интегрирование
Accuracy	Aniqlik	Точность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The trapezium rule replaces the curve by:

flat tops

straight chords

parabolas

circles

n strips need how many ordinates?

n

$n - 1$

$n + 1$

$2n$

Interior ordinates are counted:

once

twice

three times

not at all

For a concave-up curve the estimate is:

exact

too large

too small

unpredictable

Doubling the strips divides the error by about:

2

4

8

10

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 4 strips need how many ordinates?

ANSWER 5

2. 6 strips on $[0, 3]$: find h

ANSWER 0.5

3. 5 strips on $[1, 6]$: find h

ANSWER 1

4. Interior ordinates are counted

ANSWER twice

5. End ordinates are counted

ANSWER once

6. $y = x^2$ is concave

ANSWER up

7. So the estimate is

ANSWER too large

Medium · the standard question

7 PROBLEMS

1. $\int_0^4 x^2 dx$ with 4 strips

ANSWER 22

2. Its exact value

ANSWER $\frac{64}{3} \approx 21.33$

3. $\int_0^4 x^2 dx$ with 2 strips

ANSWER 24

4. $\int_1^3 \frac{1}{x} dx$ with 4 strips

ANSWER ≈ 1.117

5. Speeds 0, 8, 14, 18, 20 every 10 s: the distance

ANSWER 500 m

6. $\int_0^1 \sqrt{1+x^3} dx$ with 4 strips

ANSWER ≈ 1.111

7. Is that an over- or under-estimate?

ANSWER over — the curve is concave up

Hard · stretch and reason

7 PROBLEMS

1. $\int_0^2 \frac{1}{1+x^2} dx$ with 4 strips

ANSWER ≈ 1.1032

2. Its exact value

ANSWER $\arctan 2 \approx 1.1071$

3. Why is that one an under-estimate?

ANSWER The curve is concave down over most of the range

4. $\int_0^4 x^2 dx$ with 8 strips, and the error

ANSWER 21.5, error 0.17

5. Show the error is about h^2 times a constant

ANSWER Each doubling quarters it

6. Widths 0, 2.4, 3.1, 3.6, 3.9, 4.0 m every 5 m: the area

ANSWER $\approx 65.5 \text{ m}^2$

7. A curve is concave up then down. What can be said about the estimate?

ANSWER Nothing in general — split the interval

07 Homework

Homework — 5 tasks

Always state the number of strips, the value of h , and whether the estimate is over or under.

1. Estimate $\int_0^3 x^3 dx$ with 3 strips and compare with the exact value.
2. Estimate $\int_1^5 \frac{1}{x} dx$ with 4 strips, and say whether it is an over- or an under-estimate.
3. Estimate $\int_0^2 \sqrt{1+x^3} dx$ with 4 strips.
4. A river's depth every 4 m across is 0, 1.8, 2.6, 3.0, 2.4, 0 m. Estimate the cross-sectional area.

5. Explain in three sentences why doubling the number of strips divides the error by about four.